

Hacettepe University | Computer Engineering CMP 720 EMBEDDED SYSTEMS DESIGN

Spring 2026

Midterm Project Peer Evaluation Form

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Instructions

For each project, briefly summarize the work and provide your evaluation **individually (not as a group)**. Your response should reflect your understanding of the problem, methodology, and evaluation plan, and include critical insights, strengths, limitations, and possible improvements. Skip your own group's evaluation and simply state that it is your group.

Students are required to evaluate only the projects presented within their own class group (Hacettepe or ASELSAN).

Guiding points (address in one coherent paragraph):

- What problem is being solved, why is it important, and what are the most important embedded system design considerations for this problem (e.g., timing, reliability, energy, safety, security, scalability, etc.)?
- What is the main idea of the proposed methodology?
- What do you think are the strengths and limitations of the approach?
- Is there anything you would improve or do differently?
- What is the most critical or challenging component?
- Is the evaluation plan sufficient? What, if anything, is missing?

Group 1

Umut Kalman, Eren Karadağ

Project Title: *Context-Aware Deadline Scheduling Under Memory Contention for Embedded Real-Time Systems*

Hacettepe group.

Group 2

Yusuf Çağrı Ögüt

Project Title: *Hybrid Cellular-LoRa Network Based Agriculture Monitoring and Control Application*

Hacettepe group.

Group 3

Yusuf Tahir Orhan

Project Title: *Implementation of DeFT: A Deadlock-Free and Fault-Tolerant Routing Algorithm*

Hacettepe group.

Group 4

Mehmet Yusuf Sezgi, Rana Elif Albayrak

Project Title: *Real-Time CAN Decoder*

Hacettepe group.

Group 5

Metin Irkıcıtal, Anıl Arda

Project Title: *Edge-AI Based Smart Access Control System*

The project targets privacy and latency issues of cloud-based face recognition by moving everything to an MCU, which is a relevant embedded problem. Key design aspects are tight memory (approximately 1 MB), real-time constraints, and RTOS scheduling. The method uses an INT8 MobileNet-based embedding pipeline with cosine similarity. A strong point is the fully offline and deterministic design, but using pre-cropped inputs limits real-world usability. I would improve it by adding lightweight on-device face detection. The most critical part is fitting the ML pipeline within strict memory and timing limits. The evaluation is reasonable but lacks robustness and real-world variability tests.

Group 6

Oğulcan Uğuroğlu, Barış Büyükyılmaz

Project Title: *Contention-Aware Duplication Scheduling for NoC Systems*

Hacettepe group.

Group 7

Ahmet Yiğitalp Demirci

Project Title: *Real-Time Energy-Constrained UAV Optimization*

The project addresses energy-aware decision making in UAVs, which is important for safe mission completion. It considers real-time constraints, safety, and accurate power estimation. The method uses a PWM-based power model and a greedy algorithm for velocity selection. A strength is the more realistic energy modeling, while a limitation is the discretized solution space. I would improve it by adding adaptive behavior for environmental factors like wind. The most critical part is the accuracy of the power model. The evaluation is strong with simulation, but real hardware validation would improve it.

Group 8

Sami Enes Yılmaz, Kadir Emre Avcı

Project Title: *SmartAquaria System*

This project focuses on reliable monitoring/control of aquatic systems, which is important due to environmental sensitivity. Design aspects include portability, modularity, and soft real-time behavior. The system uses a layered architecture and FSM-based control. A strong point is the clean architecture and simulation support, but lack of sensor filtering is a weakness. I would improve it by adding filtering and anomaly detection. The most critical part is maintaining stable FSM behavior under noisy inputs. The evaluation is acceptable but lacks stress testing.

Group 9

Mehmethan Ayrim

Project Title: *Energy-Aware Mixed-Criticality Scheduling*

The project deals with reducing energy consumption in mixed-criticality systems without violating deadlines. It focuses on real-time constraints and DVFS-based scheduling. The DFU method dynamically adjusts frequency using runtime slack. A strength is achieving energy savings while preserving schedulability, but it assumes ideal DVFS conditions. I would improve it by considering discrete frequencies and hardware overheads. The most critical part is accurate utilization tracking. The evaluation is sufficient for simulation but lacks real-system validation.

Group 10

İrem Gül Subaşı

Project Title: *Fault Injection and Resilience Profiling of Lightweight Transformers*

The project analyzes reliability of transformer models on FPGA using fault injection, which is important for safety-critical AI. It focuses on system reliability and fault tolerance. The method injects bit-flip faults and analyzes layer-level sensitivity. A strength is identifying vulnerable components for selective protection, but only single-bit faults are considered. I would improve it by including multi-bit and permanent faults. The most critical part is maintaining correct execution while injecting faults. The evaluation is useful but limited in scope and lacks validation of mitigation strategies.

Group 11

Süleyman Ege Karakaya

Project Title: *Adaptive Data-Movement Strategy for Crypto Accelerators*

This project addresses the often-overlooked data movement bottleneck in embedded cryptographic systems. It considers CPU usage, memory constraints, and bus contention. The method formulates transfer-mode selection (PIO/DMA) as an optimization problem. A strong point is the system-level perspective, but contention modeling is simplified. I would improve it with more realistic workload modeling. The most critical part is keeping selector overhead low. The evaluation is good but could include more dynamic scenarios.

Group 12

Gökçer Erol

Project Title: *Acoustic Localization System for Search and Rescue*

Hacettepe group.

Group 13

Emre Doğan, Yavuz Selim Kızıltaş

Project Title: *Optimization of Edge AI Pipeline on TDA4VM*

The project focuses on optimizing edge AI inference under latency and power constraints. It considers heterogeneous computing, memory hierarchy, and synchronization. The method distributes tasks across ARM/DSP/accelerators and overlaps computation with data movement. A strength is effective hardware utilization, but it is highly platform-dependent. I would improve it by exploring adaptive scheduling. The most critical part is efficient data movement. The evaluation is strong but lacks thermal and long-term analysis.

Group 14

Mehmet Yusuf Bilge, Sezil Ağırbaş

Project Title: *Closed-Loop Vestibular Stimulation System*

Hacettepe group.

Group 15

Ayberk Aygün, Berkay Taşkın

Project Title: *Quadrotor Simulation on NVIDIA Jetson*

Hacettepe group.

Group 16

Yasin Arslan, Süleyman Sertaç Üst

Project Title: *Real-Time Audio Pipeline Analysis on Heterogeneous SoCs*

This is my group.

Group 17

Oğuz Kağan Yağlıoğlu, Emre Eşef Kılıçdoğan

Project Title: *Predictive Maintenance of BLDC Motors via Multi-Sensor Fusion*

Hacettepe group.

Group 18

Hüseyin Eren Doğan

Project Title: *EdgeRAG: Offline RAG on Embedded Systems*

This project aims to enable offline AI-based QA on resource-constrained systems, which is important for secure environments. It focuses on memory limits, latency, and offline operation. The method uses a sequential RAG pipeline with quantized models and vector search. A strong point is strict memory management, but latency is high. I would improve it by optimizing inference speed. The most critical part is avoiding OOM issues. The evaluation is detailed but lacks energy analysis and real hardware validation.

Group 19

Ali Bölücü

Project Title: *Spectral-Seeded NSGA-II for Many-Core Systems*

Hacettepe group.

Group 20

İlter Doğaç Dönmez

Project Title: *Autonomous Obstacle Avoidance Vehicle using LiDAR*

Hacettepe group.